Introduction to Electronics

TUTORIAL: ZENER DIODE

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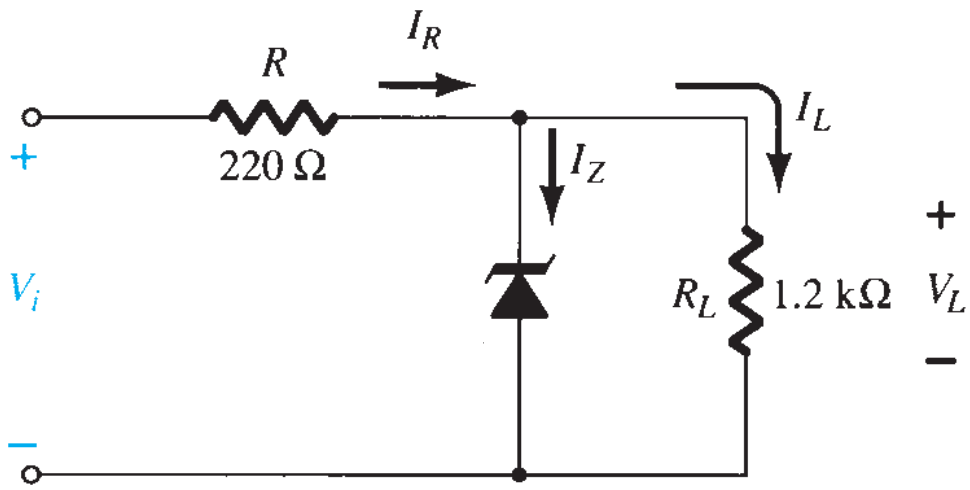
Problems

Q1: A Zener diode whose nominal voltage is 10V at 10 mA has an incremental resistance of 50 Ω.What voltage do you expect if the diode current is halved? Doubled? What is the value of V_{Z0} in the zener model?

Q2: A shunt regulator utilizes a zener diode whose voltage is 5.1 V at a current of 50 mA and whose incremental resistance is 7 Ω. The diode is fed from a supply of 15-V nominal voltage through a 200-Ω resistor. What is the output voltage at no load? Find the line regulation and the load regulation.

Problems

Q3: Determine the range of values of V_i that will maintain the Zener diode in "on" state. The maximum allowed reverse biased current for the Zener diode is 60 mA. The knee voltage is around 20V and knee current is 0.2mA and r_z=7 Ω.